ThinkJEPA: Empowering Latent World Models with Large Vision-Language Reasoning Model

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Research Questions

- How can dense-frame JEPA dynamics modeling be combined with long-horizon VLM semantic reasoning for improved latent world modeling?
- How should multi-layer VLM representations be extracted and injected into JEPA predictors to preserve both visual reasoning cues and semantic guidance?
- Can VLM-guided JEPA models achieve better long-horizon rollout stability compared to standalone JEPA or VLM approaches?

Core Idea

JEPA-style latent world models excel at fine-grained dynamics prediction but suffer from limited temporal context and weak semantic grounding. VLMs provide rich semantic reasoning over long horizons but are ill-suited for dense prediction due to sparse sampling and language output bottlenecks. ThinkJEPA solves this by using a dual-temporal design: a dense JEPA branch captures high-frequency dynamics while a uniformly-sampled VLM 'thinker' branch provides semantic guidance. The key insight is injecting hierarchical VLM representations (from multiple transformer layers) into the JEPA predictor via FiLM modulation, combining the strengths of both approaches without their individual weaknesses.

Key Formula

Uniform sampling for VLM branch

$$v_u = \left\{ I_{s_i} \right\}_{i=1}^{N_u}, \quad s_i = \left\lfloor 1 + (i-1) \cdot \frac{N-1}{N_u - 1} \right\rfloor$$

v_u is uniformly sampled clip, N_u is number of sampled frames for VLM thinker, s_i are frame indices spanning entire clip to provide large temporal perception field under limited compute

VLM-guided JEPA prediction

$$\hat{F}^{\text{fut}} = g(F^{\text{past}}(v_d); \phi(v_u), p)$$

F^{past}(v_d) are past latent tokens from dense clip, φ(v_u) are VLM-derived guidance features from uniform clip, p is text prompt, g is V-JEPA predictor conditioned on both dense dynamics and VLM guidance

Key Figure

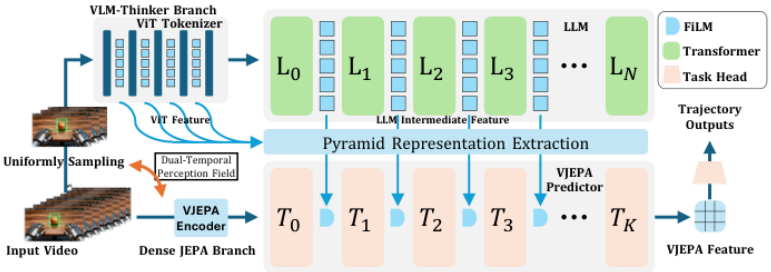

Fig. 1. Overall Architecture of ThinkJEPA. ThinkJEPA couples a dense JEPA branch for fine-grained latent dynamics modeling with a uniformly sampled VLM-*thinker* branch that provides long-horizon semantic guidance. The VLM guidance—including visual tokens from the ViT visual tokenizer and intermediate hidden states from the language model—is distilled by a *pyramidal representation extraction* module and injected into the V-JEPA predictor via layer-wise modulation. Concretely, guidance derived from language-model layers $\{L_0, \ldots, L_N\}$ is mapped to modulation parameters for predictor layers $\{T_0, \ldots, T_K\}$. The predicted future latents are concatenated with past teacher latents to form the full latent sequence, which is then fed into a task head to produce downstream trajectory predictions.

VLMs. Recent work has explored combining language models with JEPA-style representations, but largely in directions that differ from latent world modeling. For example, VL-JEPA [6] incorporates language signals into a joint-embedding predictive framework, and other approaches use V-JEPA representations as inputs to large language models for video understanding [4]. While effective for multimodal understanding, these designs often shift the primary output interface toward language generation or do not explicitly maintain a latent forecasting interface for downstream world-model tasks. In contrast, ThinkJEPA retains JEPA-style latent forecasting and leverages VLM semantics as *guidance* by injecting VLM-derived features into the JEPA predictor, preserving dense latent prediction while adding long-horizon semantic cues.

3 Methodology

3.1 Problem Definition

Basic Settings Given a video clip v with N frames, our goal is to forecast future latent representations that support downstream tasks; in this work, we focus on 3D hand trajectory prediction. We adopt a JEPA-style latent world modeling paradigm: a visual backbone encodes video frames into latent tokens, and a transformer predictor forecasts future latent tokens from past observations. To improve semantic grounding and long-horizon reasoning, we further condition the predictor on cached features from a video VLM *thinking* model

Figure 1: Shows the complete ThinkJEPA architecture with dual-temporal pathways, hierarchical pyramid extraction, and layer-wise FiLM guidance injection mechanism.

Limitations

- Requires caching and processing both dense JEPA frames and sparse VLM frames, increasing computational overhead
- Limited to relatively short video clips due to memory constraints of the dual-branch architecture
- Hierarchical pyramid extraction relies on manually selected VLM layers rather than learned layer selection
- Evaluated primarily on hand-manipulation tasks, generalization to other domains unclear